

PRAKTIKUM BASIS DATA

MODUL 9

Scripting Procedure dan Function



Oleh:

Ahmad Dhani Ibrahim

103122400058

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

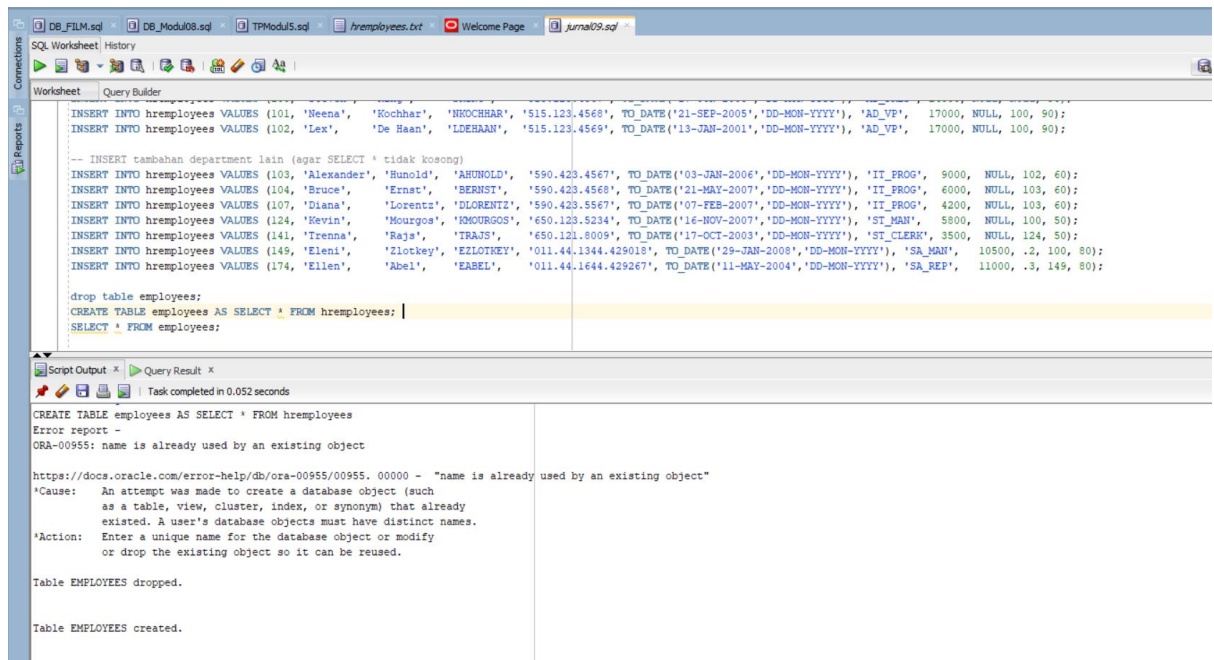
UNIVERSITAS TELKOM

2026

MODUL 9

Jurnal

1. create table employees as select * from hremployees;



The screenshot shows the SQL Developer interface. The top pane displays the SQL script for creating the 'employees' table. The script includes several INSERT statements for the 'hremployees' table and a final statement: `CREATE TABLE employees AS SELECT * FROM hremployees;`. Below the script, the 'Script Output' pane shows an error message: `ORA-00955: name is already used by an existing object`. The error report indicates that the table 'EMPLOYEES' already exists and provides instructions on how to resolve the issue.

```
INSERT INTO hremployees VALUES (101, 'Neena', 'Kochhar', 'NKOCHHAR', '515.123.4568', TO_DATE('21-SEP-2005', 'DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);
INSERT INTO hremployees VALUES (102, 'Lex', 'De Haan', 'LDEHAAN', '515.123.4569', TO_DATE('13-JAN-2001', 'DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);

-- INSERT tambahan department lain (agar SELECT * tidak kosong)
INSERT INTO hremployees VALUES (103, 'Alexander', 'Hunold', 'AHUNOLD', '590.423.4567', TO_DATE('03-JAN-2006', 'DD-MON-YYYY'), 'IT_PROG', 9000, NULL, 102, 60);
INSERT INTO hremployees VALUES (104, 'Bruce', 'Ernst', 'BERNST', '590.423.4568', TO_DATE('21-MAY-2007', 'DD-MON-YYYY'), 'IT_PROG', 6000, NULL, 103, 60);
INSERT INTO hremployees VALUES (107, 'Diana', 'Lorentz', 'DLORENTZ', '590.423.5567', TO_DATE('07-FEB-2007', 'DD-MON-YYYY'), 'IT_PROG', 4200, NULL, 103, 60);
INSERT INTO hremployees VALUES (124, 'Kevin', 'Mourgos', 'KMOURGOS', '650.123.5234', TO_DATE('16-NOV-2007', 'DD-MON-YYYY'), 'ST_MAN', 5800, NULL, 100, 50);
INSERT INTO hremployees VALUES (141, 'Trenna', 'Rajs', 'TRAJS', '650.121.8009', TO_DATE('17-OCT-2003', 'DD-MON-YYYY'), 'ST_CLERK', 3500, NULL, 124, 50);
INSERT INTO hremployees VALUES (149, 'Eleni', 'Zlotkey', 'EZLOTKEY', '011.44.1344.429018', TO_DATE('29-JAN-2008', 'DD-MON-YYYY'), 'SA_MAN', 10500, .2, 100, 80);
INSERT INTO hremployees VALUES (174, 'Ellen', 'Abel', 'EABEL', '011.44.1644.429267', TO_DATE('11-MAY-2004', 'DD-MON-YYYY'), 'SA_REP', 11000, .3, 149, 80);

drop table employees;
CREATE TABLE employees AS SELECT * FROM hremployees;
SELECT * FROM employees;
```

Script Output x Query Result x
Task completed in 0.052 seconds
CREATE TABLE employees AS SELECT * FROM hremployees
Error report -
ORA-00955: name is already used by an existing object
<https://docs.oracle.com/error-help/db/ora-00955/00955.00000> - "name is already used by an existing object"
*Cause: An attempt was made to create a database object (such as a table, view, cluster, index, or synonym) that already existed. A user's database objects must have distinct names.
*Action: Enter a unique name for the database object or modify or drop the existing object so it can be reused.
Table EMPLOYEES dropped.
Table EMPLOYEES created.

2. select * from employees;

Script Output x Query Result x

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-01-2006	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-05-2007	IT_PROG	6000	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-02-2007	IT_PROG	4200	(null)	103	60
7	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-11-2007	ST_MAN	5800	(null)	100	50
8	141	Trenna	Rajs	TRAJS	650.121.8009	17-10-2003	ST_CLERK	3500	(null)	124	50
9	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-01-2008	SA_MAN	10500	0,2	100	80
10	174	Ellen	Abel	EABEL	011.44.1644.429267	11-05-2004	SA_REP	11000	0,3	149	80

3. CREATE OR REPLACE PROCEDURE increase_salary (
department_id_in IN employees.department_id%TYPE,
increase_pct_in IN NUMBER)
IS
BEGIN
FOR employee_rec
IN (SELECT employee_id

```

        FROM employees
        WHERE department_id = increase_salary.department_id_in)
    LOOP
        UPDATE employees emp
            SET emp.salary = emp.salary + emp.salary *
increase_salary.increase_pct_in
        WHERE emp.employee_id = employee_rec.employee_id;
        DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;

```

The screenshot displays an SQL Worksheet application. The top toolbar includes icons for running queries, saving, and other standard database operations. Below the toolbar, there are two tabs: 'Worksheet' and 'Query Builder'. The 'Worksheet' tab is active, showing a PL/SQL script. The script includes a procedure named 'increase_salary' that takes 'department_id_in' and 'increase_pct_in' as parameters. It uses a loop to iterate over employees in a specific department and update their salaries by a given percentage. The script also includes a comment about dropping and recreating the 'employees' table. The bottom section of the window shows the 'Script Output' tab, which contains the execution results. The results indicate that the table 'EMPLOYEES' was successfully dropped and recreated, and the procedure 'INCREASE_SALARY' was compiled without errors. The task was completed in 0.839 seconds.

```

drop table employees;
CREATE TABLE employees AS SELECT * FROM hremployees;
SELECT * FROM employees;

CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in    IN employees.department_id%TYPE,
    increase_pct_in     IN NUMBER)
IS
BEGIN
    FOR employee_rec
    IN (SELECT employee_id
        FROM employees
        WHERE department_id = increase_salary.department_id_in)
    LOOP
        UPDATE employees emp
            SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
        WHERE emp.employee_id = employee_rec.employee_id;
        DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;

```

Script Output x Query Result x

Task completed in 0.839 seconds

*Action: Enter a unique name for the database object or modify or drop the existing object so it can be reused.

Table EMPLOYEES dropped.

Table EMPLOYEES created.

Procedure INCREASE_SALARY compiled

4. BEGIN

```
increase_salary (90, .50);
```

```
ROLLBACK;
```

```
END;
```

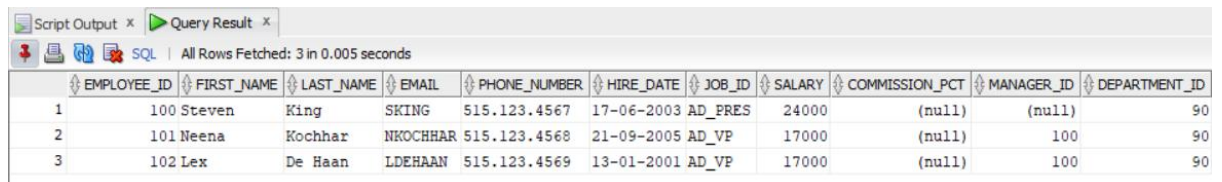
```
/
```

```
select * from employees where department_id = 90
```



EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100 Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101 Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102 Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90

5. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4.



EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100 Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101 Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102 Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90

Yang terjadi adalah perubahan kenaikan gaji akan tetap tersimpan di database, sehingga gaji pegawai pada departemen 90 jadi lebih tinggi dr awal.